

DEVICE ANATOMY 2.0

Engineering Investigation of an Electronic Device at End-of-Life

Ashtu
9/10

Course	E-Waste & Environmental Management	Class	TE INFT
Duration	45-60 minutes	Marks	10

Group Details

Group No.:	14	Device Investigated:	1) Mouse 2) Extension Board
Date:	02/09/2021	Condition:	Non-Functioned
Member 1:	Faary Sayed	Member 2:	Viraj Bhadange
Member 3:	Talha Khan	Member 4:	Darpan Phadke
Member 5:	Joey Aenoc en hurrala		
Condition: ☐ Functional ☒ Non-functional ☐ Damaged ☐ Un			



Group members actively participated in the group selfie is not considered

EACH PRINTED GROUP PHOTO



g the anatomy and end-of-life characteristics of a Extension Board
e Members

task is to investigate the material anatomy of an electronic product, identify environmental risks, and propose engineering changes that improve its end-of-life

Core question: Can you make this device more circular?

Provides multiple electrical sockets from a single power source

Device Anatomy 2.0 - E-Waste & Environmental Management - TE INFT

Part A — Device Profile

device name	Extension Board
primary function	To connect the computer wirelessly using cursor movement and buttons
Why does this device become e-waste?	☒ Obsolescence ☐ Damage ☒ Upgrade ☐ Repair not economical ☒ Failure Other: <u>Functional</u>

Part B — Engineering Anatomy

Identify at least 6 major components. Do not merely name the component — explain the material choice.

#	Component	Function	Material(s)	Why is this material suitable?
1	Outer Casing	Protects internal parts	ABS/Plastic	Strong, lightweight and provides electrical insulation
2	Copper wires	Carry electric current	Copper	Copper has excellent electrical conductivity and is flexible
3	Socket contacts	Make electrical contact with the plug	Brass/Copper	Good electrical conductivity and corrosion resistance
4	Switch	Turns the power ON/OFF	Plastic, copper, metal	Plastic provides insulation while metal contacts conduct electricity
5	Plug Pins	Connect the board to the wall power	Brass/Copper	Conduct electricity efficiently and are mechanically strong
6	Fuse	Protects supply the board from excessive current	Metal Alloy	Melts when current becomes too high, protecting the circuit

Engineering Challenge: Which component demonstrates the most interesting material selection? Why?

The most interesting material selection is copper because it is highly conductive, flexible and durable, making it suitable for carrying electricity through the extension board.

Part C — Material Detective

Tick the major material categories present in your device:

- ☒ Ferrous metals
 ☐ Aluminium
 ☒ Copper
 ☐ Precious metals
☒ Engineering plastics
☐ Glass
☐ Ceramics
☐ Semiconductor materials
☐ Rubber/elastomers
 Other: Brass

	Material 1	Material 2
Name	copper	ABS/Engineering plastic
Where is it used?	Internal wires & electrical contacts	Outer casing, switch & socket parts
Why is it used?	It conducts electricity very efficiently & is flexible.	It is strong, lightweight & electrically insulating

	Material 1	Material 2
Can it be recovered/recycled? How?	yes, The board can be dismantled & copper wires can be separated & sent for metal recycling	yes, plastic parts can be separated, sorted & recycled into new plastic products

Part D — Hidden Value: The Urban Mine

Identify the three most valuable/recoverable materials in your device.

Material	Component containing it	Potential value	Recovery difficulty
1. Copper	Internal wires & contacts	Low / Medium / High	Low / Medium / High
2. Brass	Plug pins & socket contacts	Low / Medium / High	Low / Medium / High
3. Steel / Metal	Switch & internal parts	Low / Medium / High	Low / Medium / High

Critical Thinking: Is the most valuable material necessarily the easiest to recover? Explain.

No. The value of a material does not always determine how easy it is to recover. Copper is valuable & relatively easy to separate, but materials that are tightly joined with plastic or metals can,

Part E — Hazard Detective

Identify components that may require controlled handling.

Component	Possible hazard	Risk if improperly disposed	Handling requirement
Copper wires	Electrical shock if still connected	Fire or electric shock	Disconnect power before dismantling
Fuses	may contain small metal / ceramic parts	Improper disposal add to e-waste	Send to an e-waste recycler.
Switch	electrical contacts	Fire / electric shock	Handle only when disconnected
Plastic casing	Non-biodegradable material	long-term plastic pollution	separate & recycle properly

Most environmentally concerning component: plastic casing

Why?

The plastic casing is difficult to naturally decompose & can remain in the environment for many years if it is dumped instead of being recycled.

Part F — End-of-Life Decision

Recommend the most appropriate pathway for each major component.

Component	Reuse	Repair	Refurbish	Recycle	Controlled Disposal
Plastic casing	☐	☐	☐	☒	☐
copper wires	☐	☐	☒	☒	☐
socket contacts	☐	☐	☒	☒	☐
switch	☒	☒	☐	☒	☐

Justify ONE of your decisions:

copper wires & metal contacts should be recycled to recover valuable materials. The switch should preferably be repaired or reused. Plastic should be separated & recycled.

Part G – Design for Disassembly

Imagine that you are the engineer who designed this product. Identify three design features that make recycling, repair or disassembly difficult.

#	Existing design problem
Problem 1	casing can be difficult to open without damaging it.
Problem 2	Different materials are tightly joined together.
Problem 3	Internal wires & contact are not designed for quick replacement.

Now redesign it!

Propose three engineering modifications that would make the product more circular.

Existing Design Problem	Proposed Design Change	Sustainability Benefit
Difficult casing removal	Use standard screws instead of permanent	Easier repair & disassembly
Mixed materials	Keep plastic & metal parts separately joined	Easier recycling
Difficult component replacement	Use modular replaceable switch & wiring	Extends product life & reduce waste

Sketch your redesigned device/component – show at least one change that improves repair, disassembly or material recovery.

ORIGINAL	REDESIGNED
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design &

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Part H — E-Waste Journey

Trace the end-of-life journey of ONE component.

Component selected: copper wire

DEVICE	CONSUMER DISCARD	COLLECTION	DISMANTLING	MATERIAL RECOVERY	NEW PRODUCT / RESOURCE
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What could go wrong at any stage?

The extension board may be thrown into normal waste. During dismantling, improper handling can cause electric shock or fire. If recycling is not done properly, valuable copper may be lost & materials may pollute the environment.

Part I — Circular Economy Decision

Our device should NOT simply be treated as waste because...

An extension board contains reusable parts & valuable materials such as copper, brass & plastic. These materials can be repaired, reused or recycled instead of being sent to landfill.

Which circular strategy would have the greatest impact?

☒ Repair ☒ Reuse ☐ Refurbishment ☐ Remanufacturing

☒ Recycling ☒ Material recovery

Why?

Repair & reuse can extend the life of the extension board. While recycling can recover valuable copper metal parts when the board can no longer be repaired.

part J — Engineering Viva

Be prepared to defend your answers. Your instructor may ask any of the following:

1. Why was a particular material selected for one component? $\Rightarrow$ Copper
2. Which material has the greatest recovery potential? $\Rightarrow$ Copper
3. Which component is most difficult to recycle and why? $\Rightarrow$ The switch
4. What is the single most effective design modification you propose? $\Rightarrow$ Base Board
5. If your sustainable redesign increases manufacturing cost by 8%, would you still recommend it? Why?

Because the small increase in cost can extend the product life & can improve the process of repairing & recycling

Final Group Verdict

The most valuable component is:	Copperwires
The component with the greatest environmental concern is:	Plastic casing
The biggest design-for-recycling problem is:	Difficult disassembly & separation of materials.
Our most important redesign recommendation is:	Use a screw-on casing & modular replaceable components.
ONE-SENTENCE ENGINEERING VERDICT We recommend ^{redesigning the extension board} for easy disassembly & repair because it extends product life & makes copper, metal & plastic recovery easier.	

Assessment Rubric — 10 Marks

Criterion	Excellent (2)	Good (1.5)	Developing (1)	Marks
1. Engineering Anatomy	Accurate component + function + material analysis	Mostly accurate	Basic identification	/2
2. Material & Recovery Analysis	Strong material/recovery reasoning	Adequate reasoning	Limited reasoning	/2
3. Hazard & E-waste Assessment	Identifies risks with clear justification	Identifies major risks	Basic identification	/2
4. Design for Disassembly	Innovative, feasible engineering solutions	Relevant solutions	Basic suggestions	/2
5. Engineering Defence/Viva	Clear, evidence-based justification	Reasonable explanation	Limited justification	/2
TOTAL				10

Reflection

What surprised you most about the materials hidden inside your device?

I was surprised that a simple extension board contains valuable materials such as copper & brass along with different type of plastic & metal.

After this activity, would you still call this device "waste"? Why?

No, I would not simply call it waste because the small increases in cost can extend the product life, make repair easier & improve recycling.

KEY TAKEAWAY

E-waste is not merely waste — it is a complex mixture of materials, components, resources and risks that requires engineering solutions for responsible recovery.